
CAS MA394 APPLIED ABSTRACT ALGEBRA

Spring 2026

Frank Yang

Professor Tom Enkosky

TTh 2:00 – 3:15

Contents

1	Lecture 1 – 1/20	5
1.1	1.2 Sets and Equivalence Relations	5
2	Lecture 2 – 1/21	6
3	Lecture 3 – 1/22	7
4	Lecture 4 – 1/27	8
4.1	2.1 Mathematical Induction	8
4.2	2.2 The Division Algorithm	9
5	Lecture 5 – 1/28	10
5.1	3.1 Integer Equivalence Classes and Symmetries	10
6	Lecture 6 – 1/29	11
6.1	3.2 Definitions and Examples	11
7	Lecture 7 – 2/3	12
8	Lecture 8 – 2/5	12
8.1	3.3 Subgroups	12
8.2	4.1 Cyclic Subgroups	13
9	Lecture 9 – 2/10	13
10	Lecture 10 – 2/11	13
11	Lecture 11 – 2/12	13
11.1	4.2 Multiplicative Group of Complex Numbers	13
11.2	5.1 Definitions and Notation	14
12	Lecture 12 – 2/18	14
13	Lecture 13 – 2/19	14
13.1	5.2 Dihedral Groups	14
13.2	6.1 Cosets	15
14	Lecture 14 – 2/26	15
14.1	6.2 Lagrange’s Theorem	15
14.2	6.3 Fermat’s and Euler’s Theorems	16
15	Lecture 15 – 3/3	16
15.1	9.1 Isomorphisms: Definition and Examples	16

16 Lecture 16 – 3/4	16
16.1 9.2 Direct Products	17
17 Lecture 17 – 3/5	17
18 Lecture 17 – 3/17	17
18.1 10.1 Factor Groups and Normal Subgroups	17
19 Lecture 18 – 3/18	19
19.1 11.1 Group Homomorphisms	19
20 Lecture 19 – 3/19	19
20.1 11.2 The Isomorphism Theorems	19
21 Lecture 20 – 3/24	20
22 Lecture 21 – 3/25	21
22.1 13.1 Finite Abelian Groups	21
22.2 16.1 Rings	21
23 Lecture 22 – 3/31	22
23.1 16.2 Integral Domains and Fields	22
23.2 16.3 Ring Homomorphisms and Ideals	22
24 Lecture 23 – 4/1	23
25 Lecture 24 – 4/2	23
25.1 17.1 Polynomial Rings	24
26 Lecture 25 – 4/7	25
26.1 17.2 The Division Algorithm	25
26.2 17.3 Irreducible Polynomials	25
27 Lecture 26 – 4/8	26
28 Lecture 27 – 4/9	27
28.1 18.1 Fields of Fractions	27
29 Lecture 28 – 4/16	27
29.1 20.1 Vector Spaces: Definitions and Examples	27
29.2 20.2 Subspaces	28
29.3 20.3 Linear Independence	29
30 Lecture 29 – 4/21	29
30.1 21.1 Extension Fields	29
30.2 21.2 Splitting Fields	30

31 Lecture 30 – 4/23	30
31.1 21.3 Geometric Constructions	30
32 Lecture 31 – 4/28	31
32.1 22.1 Structure of Finite Fields	31

1 Lecture 1 – 1/20

1.1 1.2 Sets and Equivalence Relations

Definition 1.1. A **set** is a well-defined collection of objects; that is, it is defined in such a manner that we can determine for any given object whether or not it belongs to the set. The objects that belong to a set are called its **elements** or **members**. We denote them with capital letters such as A or X .

- If a is an element of a set A , we write $a \in A$.
- The set without any elements is the **empty set**, denoted $\emptyset$ or $\{\}$.
- If A and B are sets and every element of A is in B , then A is a **subset** of B , denoted $A \subseteq B$.
- If $A \subset B$ and $B \subset A$, then $A = B$.
- If $A \subset B$ and $A \neq B$, then A is a **proper subset** of B , denoted $A \subset B$.

Some standard sets include:

- $\mathbb{N} = \{1, 2, 3, \dots\}$, the set of natural numbers (positive integers).
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers.
- $\mathbb{Q} = \left\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\right\}$, the set of rational numbers.
- $\mathbb{R} = (-\infty, \infty)$, the set of real numbers.
- $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$, the set of complex numbers.

We can also build new sets from old sets.

- The **union** of two sets A and B is the set of elements that are in A or B (or both), denoted $A \cup B$.

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

- The **intersection** of two sets A and B is the set of elements that are in both A and B , denoted $A \cap B$.

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

- The **complement** of a set A (with respect to a universal set U) is the set of elements in U that are not in A , denoted A' , A^c , $\bar{A}$.

$$A' = \{x : x \in U \text{ and } x \notin A\}.$$

- The **difference** of two sets A and B is the set of elements that are in A but not in B , denoted

$A \setminus B$ or $A - B$.

$$A \setminus B = A \cap B' = \{x : x \in A \text{ and } x \notin B\}.$$

Definition 1.2. De Morgan's Laws state that for any two sets A and B ,

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'.$$

Definition 1.3. The **Cartesian product** of two sets A and B is the set of ordered pairs (a, b) where $a \in A$ and $b \in B$, denoted $A \times B$.

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

An example of a Cartesian product is $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) : x, y \in \mathbb{R}\}$, the set of all points in the plane.

Definition 1.4. A **relation** R from a set A to a set B is a subset of the Cartesian product $A \times B$.

Example 1.5. Let $A = \{0, 1, 2, 3\}$, $B = \{0, 1, 2, 3\}$. Then $A \times B = \{(a, b) : a, b \in A\}$. If we say that R is the set $\{(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)\} \subset A \times B$, then R is a relation from A to B . The relation R can be interpreted as "is less than", since for each $(a, b) \in R$, we have $a < b$.

Definition 1.6. We define a mapping or function $f \subset A \times B$ from a set A to a set B as a special type of relation where each $a \in A$ has a unique $b \in B$ such that $(a, b) \in f$. We denote this by $f : A \rightarrow B$ or $A \xrightarrow{f} B$. The set A is called the **domain** of f and $f(A) = \{f(a) : a \in A\} \subset B$ is called the range or **image** of f .

2 Lecture 2 – 1/21

Definition 2.1. If $f : A \rightarrow B$ is a map and the image of f is B , i.e., $f(A) = B$, then f is called **onto** or **surjective**. In other words, if there exists an $a \in A$ for every $b \in B$ such that $f(a) = b$, then f is onto.

Definition 2.2. A map is **one-to-one** or **injective** if $a_1 \neq a_2$ implies $f(a_1) \neq f(a_2)$. Equivalently, a function is one-to-one if $f(a_1) = f(a_2)$ implies $a_1 = a_2$.

Definition 2.3. A map $f : A \rightarrow B$ is a **bijection** if it is both one-to-one and onto.

Definition 2.4. Given two functions, we can construct a new function by using the range of the first function as the domain of the second function. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be mappings. Define a new map $g \circ f : A \rightarrow C$ called the **composition** of f and g from A to C , by $g \circ f(x) = g(f(x))$.

- If S is any set, we use id_S or id to denote the **identity mapping** from S to itself, or $id(s) = s$ for all $s \in S$.
- A map $g : B \rightarrow A$ is an **inverse mapping** of $f : A \rightarrow B$ if $g \circ f = id_A$ and $f \circ g = id_B$; in other words, the inverse function of a function simply "undoes" the function.
- A map is said to be **invertible** if it has an inverse. We write f^{-1} to denote the inverse of f .

3 Lecture 3 – 1/22

Definition 3.1. An **equivalence relation** on a set X is a relation $R \subset X \times X$ such that

- $(x, x) \in R$ for all $x \in X$ (reflexive property),
- $(x, y) \in R$ implies $(y, x) \in R$ (symmetric property),
- $(x, y) \in R$ and $(y, z) \in R$ imply $(x, z) \in R$ (transitive property).

Given an equivalence relation R on a set X , we usually write $x \sim y$ instead of $(x, y) \in R$. If the equivalence relation already has an associated notation such as $=, \cong, \equiv$, we use that notation instead.

Example 3.2. Let $X = \{1, 2, 3, 4\}$. A set R that is reflexive, symmetric, and transitive (and thus an equivalence relation) is $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$.

Definition 3.3. A **partition** P of a set X is a collection of nonempty sets $X_1, X_2, \dots, X_n$ such that $X_i \cap X_j = \emptyset$ and $\bigcup_i X_i = X$. Let $\sim$ be an equivalence relation on a set X and let $x \in X$. Then $[x] = \{y \in X : x \sim y\}$ is called the **equivalence class** of x .

Example 3.4. Let $\mathbb{Z}$ be partitioned into $A = \{\text{even integers}\}$ and $B = \{\text{odd integers}\}$. $A \cap B = \emptyset$ and $A \cup B = \mathbb{Z} \implies A, B$ partition the integers. Define a relation $\sim$ on $\mathbb{Z}$ where $x \sim y \iff x, y$ both even or both odd. Every $a \in A$ is equivalent to every other $a' \in A$ and every $b \in B$ is equivalent to every other $b' \in B$.

4 Lecture 4 – 1/27

4.1 2.1 Mathematical Induction

Definition 4.1. First Principle of Mathematical Induction: Let $S(n)$ be a statement about integers for $n \in \mathbb{N}$ and suppose $S(n_0)$ is true for some integer $n_0 \in \mathbb{N}$. If for all integers k with $k \geq n_0$, $S(k)$ implies that $S(k+1)$ is true, then $S(n)$ is true for all integers $n \geq n_0$.

Example 4.2. For all $n \in \mathbb{N}$, $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

The base case is $n = 1$; then $1 = \frac{1(1+1)}{2} = 1$, so the base case holds.

Inductive hypothesis: Assume $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ for some n .

Inductive step: We want to show that the statement is true for $n+1$. Consider $1 + 2 + 3 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = (n+1)(\frac{n}{2} + 1) = (n+1)(\frac{n+2}{2}) = \frac{(n+1)(n+1+1)}{2}$.

Example 4.3. Prove that

$$\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for $n \in \mathbb{N}$.

Base: $n = 1$, then $\frac{1}{1 \cdot 2} = \frac{1}{2}$, so the base case holds.

Inductive hypothesis: Assume $\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$ for some n .

Inductive step: We want to show that the statement is true for $n+1$. Consider

$$\begin{aligned} \frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} &= \frac{n}{n+1} + \frac{1}{(n+1)(n+2)} \\ &= \frac{n(n+2) + 1}{(n+1)(n+2)} \\ &= \frac{(n+1)^2}{(n+1)(n+2)} \\ &= \frac{n+1}{n+2}. \end{aligned}$$

Therefore the statement is true for all n .

4.2 2.2 The Division Algorithm

Definition 4.4. Let a and b be integers with $b > 0$. Then there exist unique integers q and r such that $a = bq + r$ where $0 \leq r < b$.

- Let a and b be integers. If $b = ak$ for some integer k , we write $a|b$.
- An integer d is called a **common divisor** of a and b if $d|a$ and $d|b$.
- The **greatest common divisor** of a and b is a positive integer d such that d is a common divisor of a and b , and if d' is any other common divisor of a and b , then $d'|d$. We write $d = \gcd(a, b)$.
- We say that two integers a and b are relatively prime if $\gcd(a, b) = 1$.

Proof of the $\gcd$ theorem uses the division algorithm:

$$\begin{aligned}
 a &= q_1b + r_1, 0 \leq r_1 < b \\
 b &= q_2r_1 + r_2, 0 \leq r_2 < r_1, r_1 \neq 0 \\
 r_1 &= q_3r_2 + r_3, 0 \leq r_3 < r_2, r_2 \neq 0 \\
 &\vdots \\
 r_{n-2} &= q_nr_{n-1} + r_n, 0 \leq r_n < r_{n-1}, r_{n-1} \neq 0 \\
 r_{n-1} &= q_{n+1}r_n + 0.
 \end{aligned}$$

The first non-zero remainder is $r_n|\gcd(a, b)$, or a common divisor. To show that it's the greatest common divisor, suppose $d = as' + bt'$ for some $s', t' \in \mathbb{Z}$ where $d = \gcd(a, b)$. Then if $a = q_1b + r_1$ and $b = q_2r_1 + 0$, if $d > r_1$, but d is a common divisor, then $a - q_1b = r_1$. Thus $d|a, d|b \implies d|r_1$, therefor d can't be bigger than r_1 .

Example 4.5. Let $a = 23, b = 3$. Since $23 = 1 \cdot 3 + 20$, we have $q = 1$ and $r = 20$ which is not valid since r is not less than b . We also have $23 = 8 \cdot 3 - 1$, so $q = 8$ and $r = -1$ is not valid since r is not nonnegative.

The only valid one is $23 = 3 \cdot 7 + 2$, so $q = 7$ and $r = 2$. There is no other way to write $23 = 3 \cdot q + r$ where $0 \leq r < 3$.

Example 4.6. Use the Division Algorithm to find $\gcd(2520, 378)$.

$$\begin{aligned}
 2520 &= 378(6) + 252 \\
 378 &= 252(1) + 126 \\
 252 &= 126(2) + 0
 \end{aligned}$$

The last non-zero remainder is the gcd, so $\gcd(2520, 378) = 126$.

5 Lecture 5 – 1/28

Theorem 5.1. Let p be an integer such that $p > 1$. We say that p is a **prime number**, or simply p is **prime**, if the only positive divisors of p are 1 and p itself. An integer greater than 1 that is not prime is called a **composite number**.

Lemma. Euclid. Let a and b be integers and p be a prime number. If $p|ab$, then either $p|a$ or $p|b$. Suppose $p|ab$ and suppose $p \nmid a$. Then $\gcd(a, p) = 1$. Thus there exist integers s, t such that $as + pt = 1$. Multiplying both sides by b gives $abs + ptb = b$. Since $p|ab$ and $p|ptb$, we have $p|b$.

Theorem 5.2. There exist an infinite number of primes.

Assume $p_1, p_2, \dots, p_n$ are all the primes. Let $N = p_1 p_2 \cdots p_n + 1$. Then N is either prime or composite. If N is prime, then there exists a prime not in our list. If N is composite, then it has a prime divisor p . Since p divides N and p divides $p_1 p_2 \cdots p_n$, p must also divide their difference, which is 1. This is a contradiction since no prime divides 1. Thus there exists a prime not in our list.

5.1 3.1 Integer Equivalence Classes and Symmetries

$\mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}$ where arithmetic is mod n and the equivalence classes are $a \equiv b \pmod{n}$ if $n|a - b$.

Example 5.3. $\mathbb{Z}_6$:

$[0] = \{0, \pm 6, \pm 12, \pm 18 \cdots\} = [6] = [-12]$ all have a remainder of 0 when divided by 6.

$[1] = \{1, \pm 5, \pm 11, \pm 17 \cdots\} = [7] = [-11]$ all have a remainder of 1 when divided by 6.

$\vdots$

6 Lecture 6 – 1/29

Definition 6.1. Let $\mathbb{Z}_n$ be the set of equivalence classes of the integers *mod* n and $a, b, c \in \mathbb{Z}_n$.

1. Addition and multiplication are commutative. $ab \equiv ba$ and $a + b \equiv b + a$.
2. Addition and multiplication are associate. $(ab)c \equiv a(bc)$ and $(a + b) + c \equiv a + (b + c)$.
3. There are both additive and multiplicative identities. $0 \in \mathbb{Z}_n$ has the property $a + 0 \equiv a$ and $1 \in \mathbb{Z}_n$ has the property $a \cdot 1 \equiv a$.
4. Multiplication distributes over addition. $a(b + c) \equiv ab + ac$.
5. For every integer a there is an additive inverse $-a$. $a + (n - a) \equiv (n - a) + a \equiv n \equiv 0$.
6. Let a be a nonzero integer. Then $\gcd(a, n) = 1$ if and only if there exists a multiplicative b inverse for $a(\text{mod } n)$.

Definition 6.2. A **symmetry** of a geometric figure is a rearrangement of the figure preserving the arrangement of its sides and vertices as well as its distances and angles. A map from the plane to itself preserving the symmetry of an object is called a **rigid motion**.

6.1 3.2 Definitions and Examples

Definition 6.3. A **binary operation** or **law of composition** on a set G is a function $G \times G \rightarrow G$ that assigns to each pair $(a, b) \in G \times G$ a unique element $a \circ b$, or ab in G , called the composition of a and b .

Definition 6.4. A **group** is a set G together with a law of composition $\circ$ that satisfies the following axioms:

- The law of composition is **associative**; that is, for all $a, b, c \in G$, we have $(a \circ b) \circ c = a \circ (b \circ c)$.
- There exists an **identity element** e in G such that for all $a \in G$, we have $e \circ a = a \circ e = a$.
- For each element $a \in G$, there exists an **inverse element** $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = e$.

A group with the property that $a \circ b = b \circ a$ for all $a, b \in G$ is called an **abelian group** or **commutative group**. Groups not satisfying this property are called **nonabelian** or **non-commutative**.

7 Lecture 7 – 2/3

The identity element in a group G is unique; that is, there exists only one element $e \in G$ such that $eg = ge = g$ for all $g \in G$.

If g is any element in a group G , then the inverse of g , denoted by g^{-1} , is unique.

Let G be a group. If $a, b \in G$, then $(ab)^{-1} = b^{-1}a^{-1}$.

Let G be a group. For any $a \in G$, $(a^{-1})^{-1} = a$.

Let G be a group and a and b be any two elements in G . Then the equations $ax = b$ and $xa = b$ have unique solutions in G .

If G is a group and $a, b, c \in G$, then $ba = ca$ implies $b = c$ and $ab = ac$ implies $b = c$.

Theorem 7.1. In a group, the usual law of exponents hold; that is, for all $g, h \in G$,

1. $g^m g^n = g^{m+n}$ for all $m, n \in \mathbb{Z}$.
2. $(g^m)^n = g^{mn}$ for all $m, n \in \mathbb{Z}$.
3. $(gh)^n = (h^{-1}g^{-1})^{-n}$ for all $n \in \mathbb{Z}$. Furthermore, if G is abelian, then $(gh)^n = g^n h^n$.

8 Lecture 8 – 2/5

8.1 3.3 Subgroups

We define a **subgroup** H of a group G to be a subset H of G such that when the group operation of G is restricted to H , H is a group in its own right.

Observe that every group with at least two elements will always have at least two subgroups, the subgroup consisting of the identity element alone and the entire group itself. The subgroup $H = \{e\}$ of a group is called the **trivial subgroup**. A subgroup that is a proper subset of is called a **proper subgroup**.

Example 8.1. Let $G = (\mathbb{Z}, +)$, and $H = (\mathbb{Z}, +)$. H is a trivial subgroup of G . Then let $H' = (\{0\}, +)$. H' is the trivial subgroup of G .

Since $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$, we have: $(\mathbb{C}, +)$ is a group, $(\mathbb{R}, +)$ is a subgroup of $(\mathbb{C}, +)$. If $H = (\mathbb{R}, +)$ is a subgroup of $G = (\mathbb{C}, +)$, then consider $(a_1 + b_1 i) + (a_2 + b_2 i) \in G \implies (a_1 + a_2) + (b_1 + b_2)i \in G$.

Let H be a subset of group G . Then H is a subgroup of G if and only if

- **Closure:** For all $a, b \in H$, $ab \in H$.
- **Identity:** The identity element e of G is in H .
- **Inverses:** For all $a \in H$, $a^{-1} \in H$.

8.2 4.1 Cyclic Subgroups

Theorem 8.2. Let G be a group and a be any element in G . Then the set $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G . Furthermore, $\langle a \rangle$ is the smallest subgroup of G that contains a and is called the **cyclic subgroup** of G generated by a .

9 Lecture 9 – 2/10

For $a \in G$, we call $\langle a \rangle$ the **cyclic subgroup** of G generated by a .

If G contains some element a such that $G = \langle a \rangle$, then G is a **cyclic group** and a is called a **generator** of G .

If a is an element of a group G , we define the **order** of a to be the smallest positive integer n such that $a^n = e$, and we write $|a| = n$. If there is no such integer n , we say that the order of a is infinite and write $|a| = \infty$ to denote the order of a .

Theorem 9.1. Every cyclic group is abelian.

Also, every subgroup of a cyclic group is cyclic.

Corollary: The subgroups of $\mathbb{Z}$ are exactly $n\mathbb{Z}$ for $n = 0, 1, 2, \dots$.

10 Lecture 10 – 2/11

Proposition: Let G be a cyclic group of order n and suppose that a is a generator for G . Then $a^k = e$ if and only if $n|k$, or k divides n .

Theorem 10.1. Let G be a cyclic group of order n and suppose that a is a generator of the group. If $b = a^k$, then the order of b is $\frac{n}{d}$, where $d = \gcd(k, n)$.

Corollary: The generators of $\mathbb{Z}_n$ are the integers r such that $1 \leq r < n$ and $\gcd(r, n) = 1$.

11 Lecture 11 – 2/12

11.1 4.2 Multiplicative Group of Complex Numbers

$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}, i^2 = -1\}$.

If $z = x + iy$, then $z = r(\cos \theta + i \sin \theta) = cis\theta = re^{i\theta}$ where $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$.

The circle group $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ is a subgroup of $\mathbb{C}^*$.

Theorem 11.1. If $z^n = 1$, then the n^{th} roots of unity are $z = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$ for $k = 0, 1, 2, \dots, n-1$. Furthermore, the n^{th} roots of unity form a cyclic subgroup of $\mathbb{T}$ of order n .

11.2 5.1 Definitions and Notation

Theorem 11.2. The symmetric group on n letters, S_n , is a group with $n!$ elements, where the binary operation is the composition of maps.

Cycle Notation: A permutation $\sigma \in S_X$ is a cycle of length k if there exist elements $a_1, a_2, \dots, a_k \in X$ such that $\sigma(a_1) = a_2, \sigma(a_2) = a_3, \dots, \sigma(a_k) = a_1$, and $\sigma(x) = x$ for all other $x \in X$. We write $(a_1 a_2 \dots a_k)$ to denote the cycle σ .

A subgroup of S_n is called a **permutation group**.

Two cycles in S_n , $\sigma = (a_1 a_2 \dots a_k)$ and $\tau = (b_1 b_2 \dots b_m)$ are **disjoint** if $a_i \neq b_j$ for all i and j .

Let σ and τ be two disjoint cycles in S_n . Then $\sigma\tau = \tau\sigma$.

Every permutation in S_n can be written as the product of disjoint cycles.

12 Lecture 12 – 2/18

- A cycle of length 2 is called a **transposition**.
- **Proposition:** Any permutation of a finite set containing at least two elements can be written as the product of transpositions.
- If the identity id is written as the product of r transpositions, $id = \tau_1 \tau_2 \dots \tau_r$, then r is even.

Theorem 12.1. If a permutation σ can be expressed as the product of an even number of transpositions, then any other product of transpositions equaling must also contain an even number of transpositions.

We define a permutation to be even if it can be expressed as an even number of transpositions and odd if it can be expressed as an odd number of transpositions.

Theorem 12.2. The set of even permutations A_n is a subgroup of S_n .

Proposition: The number of even permutations is $S_n, n \geq 2$, is equal to the number of odd permutations; hence the order of A_n is $\frac{n!}{2}$.

13 Lecture 13 – 2/19

13.1 5.2 Dihedral Groups

For $n = 3, 4, \dots$, we define the **n th dihedral group** to be the group of rigid motions of a regular n -gon.

Theorem 13.1. The dihedral group, D_n , is a subgroup of S_n of order $2n$.

Theorem 13.2. The group $D_n, n \geq 3$, consists of all products of the elements r and s , where r has order n and s has order 2, and these two elements satisfy the relation $srs = r^{-1}$.

13.2 6.1 Cosets

Let G be a group and H be a subgroup of G . Define a **left coset** of H with representative $g \in G$ to be the set $gH = \{gh : h \in H\}$. Define a **right coset** of H with representative g to be the set $Hg = \{hg : h \in H\}$.

14 Lecture 14 – 2/26

Lemma: Let H be a subgroup of a group G and suppose that $g_1, g_2 \in G$. The following conditions are equivalent:

- $g_1H = g_2H$
- $Hg_1^{-1} = Hg_2^{-1}$
- $g_1H \subset g_2H$
- $g_2 \in g_1H$
- $g_1^{-1}g_2 \in H$.

Theorem 14.1. Let H be a subgroup of a group G . Then the left cosets of H in G form a partition of G . That is, the group G is the disjoint union of the left cosets of H in G .

14.1 6.2 Lagrange's Theorem

Proposition: Let H be a subgroup of G with $g \in G$ and define a map $\phi : H \rightarrow gH$ by $\phi(h) = gh$. The map is bijective; hence, the number of elements in H is the same as the number of elements in gH .

Theorem 14.2. Let G be a finite group and let H be a subgroup of G . Then $|G|/|H| = [G : H]$ is the number of distinct left cosets of H in G , called the **index** of H in G . In particular, the number of elements in H must divide the number of elements in G .

Corollary: Suppose that G is a finite group and $g \in G$. Then the order of g must divide the number of elements in G .

Corollary: Let $|G| = p$ with p a prime number. Then G is cyclic and any $g \in G$ such that $g \neq e$ is a generator.

Corollary: Let H and K be subgroups of a finite group G such that $K \subset H \subset G$. Then $[G : K] = [G : H][H : K]$.

14.2 6.3 Fermat's and Euler's Theorems

The **Euler ϕ -function** is the map $\phi : \mathbb{N} \rightarrow \mathbb{N}$ defined by $\phi(1) = 1$, and, for $n > 1$, $\phi(n)$ is the number of positive integers m with $1 \leq m < n$ and $\gcd(m, n) = 1$.

15 Lecture 15 – 3/3

Theorem 15.1. Let $U(n)$ be the group of units in $\mathbb{Z}_n$ (units are elements with multiplicative inverses, aka coprime to n) under multiplication. Then $|U(n)| = \phi(n)$.

Theorem 15.2. Euler's theorem: Let a and n be integers such that $n > 0$ and $\gcd(a, n) = 1$. Then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Example 15.3. What is the last digit of 457^{327} ?

Since $457 \equiv 7 \pmod{10} \implies 457^{327} \equiv 7^{327} \pmod{10}$. Since $\gcd(7, 10) = 1$, we can apply Euler's theorem to get $7^{\phi(10)} \equiv 1 \pmod{10}$. Since $\phi(10) = 4$, we have $7^4 \equiv 1 \pmod{10}$. Thus $7^{327} \equiv 7^{4 \cdot 81 + 3} \equiv (7^4)^{81} \cdot 7^3 \equiv 1^{81} \cdot 343 \equiv 3 \pmod{10}$. Therefore the last digit of 457^{327} is 3.

Theorem 15.4. Fermat's Little Theorem: Let p be any prime number and suppose that $p \nmid a$ (p does not divide a). Then $a^{p-1} \equiv 1 \pmod{p}$. Furthermore, for any integer b , $b^p \equiv b \pmod{p}$.

15.1 9.1 Isomorphisms: Definition and Examples

Two groups $(G, \cdot)$ and $(H, \circ)$ are **isomorphic** if there exists a one-to-one and onto map $\phi : G \rightarrow H$ such that the group operation is preserved; that is, $\phi(a \cdot b) = \phi(a) \circ \phi(b)$ for all $a, b \in G$. If G is isomorphic to H , we write $G \cong H$. The map is called an isomorphism.

16 Lecture 16 – 3/4

Theorem 16.1. All cyclic groups of infinite order are isomorphic to $\mathbb{Z}$.

Theorem 16.2. If G is a cyclic group of order n , then G is isomorphic to $\mathbb{Z}_n$.

Corollary: If G is a group of order p , where p is a prime number, then G is isomorphic to $\mathbb{Z}_p$.

Theorem 16.3. Cayley: Every group is isomorphic to a group of permutations.

16.1 9.2 Direct Products

If $(G, \cdot)$ and $(H, \circ)$ are groups, then we can make the Cartesian product of G and H into a new group. As a set, our group is just the ordered pairs $(g, h) \in G \times H$. We can define a binary operation on $G \times H$ by $(g_1, h_1)(g_2, h_2) = (g_1 \cdot g_2, h_1 \circ h_2)$.

Proposition: Let G and H be groups. The set $G \times H$ is a group. The group $G \times H$ is called the **external direct product** of G and H .

Theorem 16.4. Let $(g, h) \in G \times H$. If g and h have finite orders r and s respectively, then the order of (g, h) in $G \times H$ is the least common multiple of r and s .

Theorem 16.5. The group $\mathbb{Z}_m \times \mathbb{Z}_n$ is isomorphic to $\mathbb{Z}_{mn}$ if and only if $\gcd(m, n) = 1$.

17 Lecture 17 – 3/5

Corollary: Let $n_1, \dots, n_k$ be positive integers. Then

$$\prod_{i=1}^k \mathbb{Z}_{n_i} \cong \mathbb{Z}_{n_1 \cdots n_k}$$

if and only if $\gcd(n_i, n_j) = 1$ for $i \neq j$.

Corollary: If $m = p_1^{e_1} \cdots p_k^{e_k}$ where the p_i s are distinct primes, then

$$\mathbb{Z}_m \cong \mathbb{Z}_{p_1^{e_1}} \times \cdots \times \mathbb{Z}_{p_k^{e_k}}.$$

Definition 17.1. Let G be a group with subgroups H and K satisfying the following conditions:

- $G = HK = \{hk : h \in H, k \in K\}$,
- $H \cap K = \{e\}$,
- $hk = kh$ for all $h \in H$ and $k \in K$.

Then G is the **internal direct product** of H and K .

Theorem 17.2. Let G be the internal direct product of subgroups H and K . Then G is isomorphic to $H \times K$.

18 Lecture 17 – 3/17

18.1 10.1 Factor Groups and Normal Subgroups

A subgroup H of a group G is **normal** in G if $gH = Hg$ for all $g \in G$.

Example 18.1. Consider $3\mathbb{Z}$ in $\mathbb{Z}$.

$$H = 3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\} = \{3n : n \in \mathbb{Z}\}.$$

$$0 + H = 3 + H = 6 + H = \dots = H + 0 = H + 3 = \dots \quad 1 + H = 4 + H = 7 + H = \dots = H + 1 = H + 4 = \dots \quad 2 + H = 5 + H = 8 + H = \dots = H + 2 = H + 5 = \dots$$

If G is abelian, then every subgroup is normal.

Example 18.2. Consider $H = \{e, (12)\}$ in S_3 .

$$\text{Note } eH = He = H \text{ and } (12)H = \{(12), e\} = \{e, (12)\} = H(12).$$

$$\text{Consider } (123)H = \{(123)e, (123)(12)\} = \{(123), (13)\} \text{ and } H(123) = \{e(123), (12)(123)\} = \{(123), (23)\}.$$

$$H(123) = \{(123), (23)\} \neq \{(123), (13)\} = (123)H. \text{ Thus } H \text{ is not normal in } S_3.$$

Example 18.3. Consider A_3 in S_3 .

$$A_3 = \{e, (123), (132)\}. \text{ We know that } eA_3 = A_3e = A_3. \text{ Also, } (123)A_3 = \{(123), (132), e\} = A_3(123) \text{ and } (132)A_3 = \{(132), e, (123)\} = A_3(132).$$

$$\text{Consider } (12)A_3 = \{(12), (13), (23)\} \text{ and } A_3(12) = \{(12), (13), (23)\}.$$

$$\text{So } (12)A_3 = A_3(12), \text{ and } A_3 \text{ is normal in } S_3.$$

Theorem 18.4. Let G be a group and N be a subgroup of G . Then the following statements are equivalent.

- The subgroup N is normal in G .
- For all $g \in G$, $gNg^{-1} = N$.
- For all $g \in G$, $gNg^{-1} \subset N$.

Definition 18.5. If N is a normal subgroup of a group G , then the cosets of N in G form a group G/N under the operation $(aN)(bN) = abN$. This group is called the **factor** or **quotient group** of G and N .

Example 18.6. What is the factor group $\mathbb{Z}/3\mathbb{Z}$?

$$\mathbb{Z}/3\mathbb{Z} = \{0 + 3\mathbb{Z}, 1 + 3\mathbb{Z}, 2 + 3\mathbb{Z}\} =$$

$$\{\{\dots, -6, -3, 0, 3, 6, \dots\}, \{\dots, -5, -2, 1, 4, 7, \dots\}, \{\dots, -4, -1, 2, 5, 8, \dots\}\}.$$

$$(0 + 3\mathbb{Z}) + (1 + 3\mathbb{Z}) = (0 + 1) + 3\mathbb{Z} = 1 + 3\mathbb{Z}.$$

$$(1 + 3\mathbb{Z}) + (2 + 3\mathbb{Z}) = (1 + 2) + 3\mathbb{Z} = 0 + 3\mathbb{Z}.$$

$$(2 + 3\mathbb{Z}) + (2 + 3\mathbb{Z}) = (2 + 2) + 3\mathbb{Z} = 1 + 3\mathbb{Z}.$$

Theorem 18.7. Let N be a normal subgroup of a group G . The cosets of N in G form a group G/N of order $[G : N]$.

19 Lecture 18 – 3/18

19.1 11.1 Group Homomorphisms

A **homomorphism** between groups $(G, \cdot)$ and $(H, \circ)$ is a map $\phi : G \rightarrow H$ such that $\phi(g_1 \cdot g_2) = \phi(g_1) \circ \phi(g_2)$ for $g_1, g_2 \in G$. The range of ϕ is called the **homomorphic image** of G .

Example 19.1. Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_4 \times \mathbb{Z}_9$ be a homomorphism with $\phi(1) = (2, 3)$. Find $\phi(n)$ for any $n \in \mathbb{Z}$. Which elements $n \in \mathbb{Z}$ have $\phi(n) = (0, 0)$?

$$\phi(2) = \phi(1 + 1) = \phi(1) + \phi(1) = (2, 3) + (2, 3) = (0, 6) \in \mathbb{Z}_4 \times \mathbb{Z}_9.$$

$$\phi(3) = \phi(1) + \phi(1) + \phi(1) = \phi(2) + \phi(1) = (0, 6) + (2, 3) = (2, 0).$$

$$\phi(4) = (0, 3), \phi(5) = (2, 6), \phi(6) = (0, 0).$$

Let $\phi : G_1 \rightarrow G_2$ be a homomorphism of groups. Then

1. If e is the identity of G_1 , then $\phi(e)$ is the identity of G_2 .
2. For any element $g \in G$, $\phi(g^{-1}) = [\phi(g)]^{-1}$.
3. If H_1 is a subgroup of G_1 , then $\phi(H_1)$ is a subgroup of G_2 .
4. If H_2 is a subgroup of G_2 , then $\phi^{-1}(H_2) = \{g \in G_1 : \phi(g) \in H_2\}$ is a subgroup of G_1 ;
Furthermore, if H_2 is normal in G_2 , then $\phi^{-1}(H_2)$ is normal in G_1 .

Let $\phi : G \rightarrow H$ be a group homomorphism and suppose that e is the identity of H . The subgroup $\phi^{-1}(\{e\})$ is called the **kernel** of ϕ and is denoted by $\ker(\phi)$.

Theorem 19.2. Let $\phi : G \rightarrow H$ be a group homomorphism. Then the kernel of ϕ is a normal subgroup of G .

20 Lecture 19 – 3/19

20.1 11.2 The Isomorphism Theorems

Let H be a normal subgroup of G . Define the **natural** or **canonical homomorphism** $\phi : G \rightarrow G/H$ by $\phi(g) = gH$.

Example 20.1. Let $G = \mathbb{Z}$ and $H = 5\mathbb{Z}$.

Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/5\mathbb{Z}$, $\phi(n) = n + 5\mathbb{Z}$.

$$\phi(0) = 0 + 5\mathbb{Z} = [0], \phi(1) = 1 + 5\mathbb{Z} = [1], \phi(2) = 2 + 5\mathbb{Z} = [2], \phi(3) = 3 + 5\mathbb{Z} = [3],$$

$$\phi(4) = 4 + 5\mathbb{Z} = [4], \phi(5) = 0 + 5\mathbb{Z} = [0]. \text{ Note: } \ker(\phi) = 5\mathbb{Z} \text{ (kernel: a group and happens to be the image of } \phi).$$

Theorem 20.2. First Isomorphism Theorem: If $\psi : G \rightarrow H$ is a group homomorphism with $K = \ker\psi$, then K is normal in G . Let $\phi : G \rightarrow G/K$ be the canonical homomorphism. Then there exists a unique isomorphism $\eta : G/K \rightarrow \psi(G)$ such that $\psi = \eta\phi$.

Example 20.3. Let $\psi : \mathbb{Z} \rightarrow \mathbb{Z}_6$ be a group homomorphism with $\psi(1) = 2$. Verify the First Isomorphism Theorem.

$$\psi(n) = 2n \pmod{6}. K = \ker\psi = 3\mathbb{Z} \subset \mathbb{Z}.$$

Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/\ker\psi \implies \phi : \mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$ be the canonical homomorphism. Then the First Iso. Theorem $\implies \mathbb{Z}/3\mathbb{Z} \cong \text{image of } \psi$.

Theorem 20.4. Second Isomorphism Theorem: Let H be a subgroup of a group G (not necessarily normal in G) and N a normal subgroup of G . Then HN is a subgroup of G , $H \cap N$ is a normal subgroup of H , and $HN/N \cong H/H \cap N$.

Example 20.5. Let $H = 6\mathbb{Z}$ and $N = 10\mathbb{Z}$ be subgroups of $G = \mathbb{Z}$. Verify the Second Isomorphism Theorem.

$$HN = \{h + n : h \in H, n \in N\} \implies (6\mathbb{Z})(10\mathbb{Z}) = \{(6m) + (10n) : m, n \in \mathbb{Z}\} = 2\mathbb{Z}.$$

$$H \cap N = \{m : m = 6k, m = 10n \text{ for some } k, n \in \mathbb{Z}\} = 30\mathbb{Z}.$$

$$H/H \cap N = 6\mathbb{Z}/30\mathbb{Z} = \{[0], [6], [12], [18], [24]\}.$$

$$HN/N = 2\mathbb{Z}/10\mathbb{Z} = \{[0], [2], [4], [6], [8]\}.$$

21 Lecture 20 – 3/24

Theorem 21.1. Correspondence Theorem: Let N be a normal subgroup of a group G . Then $H \mapsto H/N$ is a one-to-one correspondence between the set of subgroups H of G containing N and the set of subgroups of G/N . Furthermore, the normal subgroups of G containing N correspond to normal subgroups of G/N .

Theorem 21.2. Third Isomorphism Theorem: Let G be a group and N and H be normal subgroups of G with $N \subset H$. Then $G/H \cong \frac{G/N}{H/N}$.

Example 21.3. Let $G = \mathbb{Z}$, $H = 4\mathbb{Z}$, and $N = 24\mathbb{Z}$. Verify the Third Isomorphism Theorem.

$$24\mathbb{Z} = N \subset 4\mathbb{Z} = H \subset \mathbb{Z} = G.$$

$$G/H = \mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4.$$

$$G/N = \mathbb{Z}/24\mathbb{Z} \cong \mathbb{Z}_{24}.$$

$$H/N = 4\mathbb{Z}/24\mathbb{Z} \cong \mathbb{Z}_6.$$

$$\frac{G/N}{H/N} = \frac{\mathbb{Z}/24\mathbb{Z}}{4\mathbb{Z}/24\mathbb{Z}} \cong \mathbb{Z}_4.$$

22 Lecture 21 – 3/25

22.1 13.1 Finite Abelian Groups

- Suppose that G is a group and let $\{g_i\}$ be a set of elements in G , where i is in some index set I (not necessarily finite). The smallest subgroup of containing all of the g_i 's is the subgroup of G **generated** by the g_i 's.
- If this subgroup of G is in fact all of G , then G is **generated** by the set $\{g_i : i \in I\}$. In this case the g_i 's are called **generators** of G .
- If there is a finite set $\{g_i : i \in I\}$ that generates G , then G is **finitely generated**.
- Define a group G to be a p -group if every element in G has as its order a power of p , where p is a prime number.

Theorem 22.1. Fundamental Theorem of Finite Abelian Groups: Every finite abelian group G is isomorphic to a direct product of cyclic groups of the form

$$\mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_n^{a_n}}$$

where the p_i 's are primes (not necessarily distinct).

Theorem 22.2. Fundamental Theorem of Finitely Generated Abelian Groups: Every finitely generated abelian group is isomorphic to a direct product of cyclic groups of the form

$$\mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_n^{a_n}} \times \mathbb{Z} \cdots \times \mathbb{Z}$$

where the p_i 's are primes (not necessarily distinct).

Many of the sets we have explored have 2 operations: addition and multiplication. For example, $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_n, M_2(\mathbb{R}) = \{2 \times 2 \text{ matrices in } \mathbb{R}\}$, etc. Under addition, all of these sets are abelian, and most are groups. Under multiplication, most of these sets are not groups.

22.2 16.1 Rings

A nonempty set R is a ring if it has two closed binary operations, addition and multiplication, satisfying the following conditions:

- $a + b = b + a$ for $a, b \in R$.
- $a + (b + c) = (a + b) + c$ for $a, b, c \in R$.
- There is an element $0 \in R$ such that $a + 0 = 0 + a$ for all $a \in R$.
- For every element $a \in R$, there is an element $-a \in R$ such that $a + (-a) = 0$.
- $(ab)c = a(bc)$ for all $a, b, c \in R$.

- $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ for $a, b, c \in R$.

If there is an element $1 \in R$ such that $1 \neq 0$ and $1a = a1 = a$ for each element $a \in R$, we say that R is a ring with **unity** or **identity**.

A ring R for which $ab = ba$ for all $a, b \in R$ is called a **commutative ring**.

A commutative ring with identity is called an **integral domain** if, for ever $a, b \in R$ such that $ab = 0$, either $a = 0$ or $b = 0$.

A **division ring** is a ring R , with an identity, in which every nonzero element is a unit; that is, for each $a \in R$ with $a \neq 0$, there exists a unique element $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$.

A commutative division ring is called a **field**.

23 Lecture 22 – 3/31

23.1 16.2 Integral Domains and Fields

Cancellation Law: Let D be a commutative ring with identity. Then D is an integral domain if and only if for all nonzero elements $a \in D$ with $ab = ac$, we have $b = c$.

Theorem 23.1. Every finite integral domain is a field.

Let $D = \{0, 1, a_1, a_2, \dots, a_n\}$ be a finite

For any nonnegative integer n and any element r in a ring R we write $r + r + \dots + r$ (n times) as nr . We define the **characteristic** of a ring R to be the least positive integer n such that $nr = 0$ for all $r \in R$. If no such integer exists, then the characteristic of R is defined to be 0. We will denote the characteristic of R by $\text{char}(R)$.

Lemma: Let R be a ring with identity. If 1 has order n , then the characteristic of R is n .

Theorem 23.2. The characteristic of an integral domain is either prime or zero.

Every ring is an abelian group under addition.

If R is a finite ring, then we know the structure under addition, but there may be several types of multiplication.

23.2 16.3 Ring Homomorphisms and Ideals

An **ideal** in a ring R is a subring I of R such that if a is in I and r is in R , then both ar and ra are in I ; that is $rI \subset I$ and $Ir \subset I$ for all $r \in R$.

If R and S are rings, then a **ring homomorphism** is a map $\phi : R \rightarrow S$ satisfying

$$\phi(a + b) = \phi(a) + \phi(b)$$

$$\phi(ab) = \phi(a)\phi(b)$$

for all $a, b \in R$.

If ϕ is a one-to-one and onto homomorphism, then ϕ is called an **isomorphism** of rings.

24 Lecture 23 – 4/1

Theorem 24.1. First Isomorphism Theorem for Rings: Let $\psi : R \rightarrow S$ be a ring homomorphism. Then $\ker\psi$ is an ideal of R . If $\phi : R \rightarrow R/\ker\psi$ is the canonical homomorphism, then there exists a unique isomorphism $\eta : R/\ker\psi \rightarrow \text{im}\psi$ such that $\psi = \eta\phi$.

25 Lecture 24 – 4/2

Theorem 25.1. Second Isomorphism Theorem for Rings: Let I be a subring of a ring R and let J be an ideal of R . Then $I \cap J$ is an ideal of I and $I/I \cap J \cong (I + J)/J$.

Theorem 25.2. Third Isomorphism Theorem for Rings: Let R be a ring and I and J be ideals of R where $J \subset I$. Then $R/J \cong \frac{R/J}{I/J}$.

Theorem 25.3. Correspondence Theorem for Rings: Let I be an ideal of a ring R . Then $S \mapsto S/I$ is a one-to-one correspondence between the set of subrings S containing I and the set of subrings of R/I . Furthermore, the ideals of R containing I correspond to ideals of R/I .

A proper ideal M of a ring R is a **maximal ideal** of R if the ideal M is not a proper subset of any ideal of R except R itself. That is, M is a maximal ideal if for any ideal I properly containing M , $I = R$.

Theorem 25.4. Let R be a commutative ring with identity and M an ideal in R . Then M is a maximal ideal of R if and only if R/M is a field.

25.1 17.1 Polynomial Rings

Definition 25.5. Throughout this chapter, R denotes a commutative ring with identity.

- An expression of the form

$$f(x) = \sum_{i=1}^n a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

where $a_i \in R$ and $a_n \neq 0$, is called a **polynomial over R** with **indeterminate x** .

- The elements $a_0, a_1, \dots, a_n$ are called the **coefficients** of f , and a_n is called the **leading coefficient**.
- A polynomial is **monic** if its leading coefficient is 1.
- If n is the largest nonnegative integer such that $a_n \neq 0$, we say the **degree** of f is n and write $\deg f(x) = n$.
- If no such n exists (i.e., $f = 0$ is the zero polynomial), the degree of f is defined to be $-\infty$.
- The set of all polynomials with coefficients in R is denoted $R[x]$.
- Two polynomials $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ and $q(x) = b_0 + b_1 x + \cdots + b_m x^m$ are **equal** if $a_i = b_i$ for all $i \geq 0$.

Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ and $q(x) = b_0 + b_1 x + \cdots + b_m x^m$ be polynomials in $R[x]$. We define the sum of p and q to be the polynomial

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_k + b_k)x^k$$

where $k = \max(n, m)$ and we let $a_i = 0$ for $i > n$ and $b_j = 0$ for $j > m$. We define the product of p and q to be the polynomial

$$p(x)q(x) = \sum_{i=0}^{n+m} c_i x^i$$

where $c_i = \sum_{j=0}^i a_j b_{i-j}$.

Theorem 25.6. Let R be a commutative ring with identity. Then $R[x]$ is a commutative ring with identity.

Theorem 25.7. Let $p(x)$ and $q(x)$ be polynomials in $R[x]$, where R is an integral domain. Then

$$\deg(p(x)q(x)) = \deg p(x) + \deg q(x).$$

Furthermore, $R[x]$ is an integral domain.

26 Lecture 25 – 4/7

26.1 17.2 The Division Algorithm

Theorem 26.1. Let $f(x)$ and $g(x)$ be polynomials in $F[x]$, where F is a field and $g(x)$ is a nonzero polynomial. Then there exist unique polynomials $q(x), r(x) \in F[x]$ such that $f(x) = g(x)q(x) + r(x)$, where either $\deg r(x) < \deg g(x)$ or $r(x) = 0$.

Zero/Root of a Polynomial: Let $p(x) \in F[x]$, where F is a field, and let $\phi_a :$

Theorem 26.2. Factor Theorem: Let F be a field. An element $\alpha \in F$ is a zero of $p(x) \in F[x]$ if and only if $(x - \alpha)$ is a factor of $p(x)$ in $F[x]$.

Theorem 26.3. Bound on Number of Zeros: Let F be a field. A non-zero polynomial $p(x)$ of degree n in $F[x]$ has at most n distinct zeros in F .

Greatest Common Divisor of Polynomials: Let F be a field. A monic polynomial $d(x) \in F[x]$ is a **greatest common divisor** of $p(x), q(x) \in F[x]$ if $d(x)$ evenly divides both $p(x)$ and $q(x)$, and if for any other polynomial $d'(x)$ dividing both $p(x)$ and $q(x)$, we have $d'(x)|d(x)$. We write $d(x) = \gcd(p(x), q(x))$. Two polynomials $p(x)$ and $q(x)$ are **relatively prime** if $\gcd(p(x), q(x)) = 1$.

Theorem 26.4. Existence and Uniqueness of the GCD: Let F be a field and suppose $d(x)$ is a greatest common divisor of $p(x)$ and $q(x)$ in $F[x]$. Then there exist polynomials $r(x)$ and $s(x)$ such that $d(x) = r(x)p(x) + s(x)q(x)$. Furthermore, the greatest common divisor of two polynomials is unique.

26.2 17.3 Irreducible Polynomials

A nonconstant polynomial $f(x) \in F[x]$ is **irreducible over a field F** if $f(x)$ cannot be expressed as a product $g(x)h(x)$ where $g(x), h(x) \in F[x]$ and the degrees of $g(x)$ and $h(x)$ are both strictly less than $\deg f(x)$. Irreducible polynomials function as the "prime numbers" of polynomial rings.

Theorem 26.5. Rational Coefficient Decomposition: Let $p(x) \in \mathbb{Q}[x]$. Then

$$p(x) = \frac{r}{s}(a_0 + a_1x + \cdots + a_nx^n)$$

where $r, s, a_0, \dots, a_n$ are integers, the a_i 's are relative prime, and $\gcd(r, s) = 1$.

Theorem 26.6. Gauss's Lemma: Let $p(x) \in \mathbb{Z}[x]$ be a monic polynomial such that $p(x)$ factors into a product of two polynomials $\alpha(x)$ and $\beta(x)$ in $\mathbb{Q}[x]$, where the degrees of both $\alpha(x)$ and $\beta(x)$ are less than the degree of $p(x)$. Then $p(x) = a(x)b(x)$ where $a(x)$ and $b(x)$ are monic polynomials in $\mathbb{Z}[x]$ with $\deg \alpha(x) = \deg a(x)$ and $\deg \beta(x) = \deg b(x)$.

27 Lecture 26 – 4/8

A proper ideal P in a commutative ring R is called a **prime ideal** if whenever $ab \in P$, then either $a \in P$ or $b \in P$.

Let R be a commutative ring with identity 1, where $1 \neq 0$. Then P is a prime ideal in R if and only if R/P is an integral domain.

Principal Ideal in $F[x]$: Let F be a field. Recall that a **principal ideal** in $F[x]$ is an ideal $\langle p(x) \rangle$ generated by some polynomial $p(x)$; that is, $\langle p(x) \rangle = \{p(x)q(x) : q(x) \in F[x]\}$.

Theorem 27.1. Every Ideal in $F[x]$ is Principal: If F is a field, then every ideal in $F[x]$ is a principal ideal.

Theorem 27.2. Maximal Ideals and Irreducible Polynomials: Let F be a field and $p(x) \in F[x]$. Then $\langle p(x) \rangle$ is a maximal ideal of $F[x]$ if and only if $p(x)$ is irreducible over F .

Theorem 27.3. Integer Zeros of Monic Integer Polynomials: Let $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_0 \neq 0$. If $p(x)$ has a zero in $\mathbb{Q}$, then it also has a zero $\alpha \in \mathbb{Z}$, and moreover $\alpha | a_0$.

Theorem 27.4. Eisenstein's Criterion: Let p be a prime and suppose $f(x) = a_nx^n + \cdots + a_0 \in \mathbb{Z}[x]$. If $p | a_i$ for $i = 0, 1, \dots, n-1$, but $p \nmid a_n$ and $p^2 \nmid a_0$, then $f(x)$ is irreducible over $\mathbb{Q}$.

28 Lecture 27 – 4/9

28.1 18.1 Fields of Fractions

Ordered-pair set and equivalence relation: Let D be an integral domain. Define

$$S = \{(a, b) : a, b \in D, b \neq 0\}$$

Declare a relation $\sim$ on S by

$$(a, b) \sim (c, d) \iff ad = bc.$$

Lemma: The relation $\sim$ on S is an equivalence relation.

Example 28.1. Verifying the equivalence relation in $\mathbb{Z}$: Work in $D = \mathbb{Z}$. Check that $(2, 4) \sim (3, 6)$ and $(3, 6) \sim (5, 10)$, and hence $(2, 4) \sim (5, 10)$. All three pairs represent $\frac{1}{2}$.

Operations on equivalence classes: Let D be an integral domain. Denote the equivalence class of $(a, b) \in S$ under $\sim$ by $[a, b]$, and let F_D be the set of all such equivalence classes. Define addition and multiplication on F_D by

$$[a, b] + [c, d] = [ad + bc, bd], \quad [a, b] \cdot [c, d] = [ac, bd].$$

Lemma: The set F_D , together with the operations of addition and multiplication defined above, is a field.

Theorem 28.2. Let D be an integral domain. Then D can be embedded in a field of fractions F_D , where every element of F_D can be expressed as the quotient of two elements of D . Furthermore, F_D is unique in the sense that if E is any field containing D , then there exists an injective ring homomorphism $\psi : F_D \rightarrow E$ satisfying $\psi(a) = a$ for all $a \in D$.

Corollary: Let F be a field of characteristic zero. Then F contains a subfield isomorphic to $\mathbb{Q}$.

Corollary: Let F be a field of characteristic p (prime). Then F contains a subfield isomorphic to $\mathbb{Z}_p$.

29 Lecture 28 – 4/16

29.1 20.1 Vector Spaces: Definitions and Examples

A **vector space** V over a field F is a nonempty set V together with two binary operations: vector addition $+ : V \times V \rightarrow V$ and scalar multiplication $\cdot : F \times V \rightarrow V$, satisfying the following axioms for all $\vec{u}, \vec{v}, \vec{w} \in V$ and $\alpha, \beta \in F$:

- $\vec{u} + \vec{v} \in V$ (closure under addition)
- $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (commutativity)

- $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ (associativity of addition)
- There exists a vector $\vec{0} \in V$ such that $\vec{u} + \vec{0} = \vec{u}$ for all $\vec{u} \in V$ (additive identity)
- For every $\vec{u} \in V$, there exists $-\vec{u} \in V$ such that $\vec{u} + (-\vec{u}) = \vec{0}$ (additive inverses)
- $\alpha\vec{u} \in V$ (closure under scalar multiplication)
- $\alpha(\vec{u} + \vec{v}) = \alpha\vec{u} + \alpha\vec{v}$ (distributivity over vector addition)
- $(\alpha + \beta)\vec{u} = \alpha\vec{u} + \beta\vec{u}$ (distributivity over scalar addition)
- $\alpha(\beta\vec{u}) = (\alpha\beta)\vec{u}$ (associativity of scalar multiplication)
- There exists $1 \in F$ such that $1 \cdot \vec{u} = \vec{u}$ for all $\vec{u} \in V$ (multiplicative identity)

Example 29.1. The set $\mathbb{R}^n$ of all n -tuples of real numbers, $\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R}\}$, with componentwise addition and scalar multiplication is a vector space over $\mathbb{R}$.

Example 29.2. The set $M_{m \times n}(F)$ of all $m \times n$ matrices with entries in a field F , with the usual matrix addition and scalar multiplication, is a vector space over F .

Theorem 29.3. Let V be a vector space over a field F . Then each of the following holds:

- $0 \cdot \vec{v} = \vec{0}$ for every $\vec{v} \in V$, where $0 \in F$ is the zero scalar and $\vec{0} \in V$ is the zero vector.
- $\alpha \cdot \vec{0} = \vec{0}$ for every $\alpha \in F$.
- $(-1)\vec{v} = -\vec{v}$ for every $\vec{v} \in V$.
- If $\alpha\vec{v} = \vec{0}$, then either $\alpha = 0$ or $\vec{v} = \vec{0}$.

29.2 20.2 Subspaces

Definition 29.4. Let V be a vector space over a field F and let W be a nonempty subset of V . We say that W is a **subspace** of V if W is itself a vector space under the operations of V ; equivalently, W is a subspace of V if and only if it is closed under vector addition and scalar multiplication:

- If $\vec{u}, \vec{v} \in W$, then $\vec{u} + \vec{v} \in W$.
- If $\vec{v} \in W$ and $\alpha \in F$, then $\alpha\vec{v} \in W$.

Definition 29.5. Let V be a vector space over a field F and let $v_1, v_2, \dots, v_n \in V$, $\alpha_1, \alpha_2, \dots, \alpha_n \in F$. Any vector of the form $w = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$ is called a **linear combination** of $v_1, v_2, \dots, v_n$. The set of all linear combinations of a subset $S = \{v_1, \dots, v_n\}$ of V is called the **span** of S and is denoted by $\text{span}(S)$.

Theorem 29.6. Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors in a vector space V . Then $\text{span}(S)$ is a subspace of V .

29.3 20.3 Linear Independence

Definition 29.7. Let $S = \{v_1, v_2, \dots, v_n\}$ be a set of vectors in a vector space V over a field F . We say that S is **linearly dependent** if there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_n \in F$, not all zero, such that $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = \vec{0}$. If the only solution to this equation is $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$, then S is called **linearly independent**.

A set $\{v_1, v_2, \dots, v_n\}$ of vectors in a vector space V is linearly dependent if and only if at least one of the vectors v_k can be written as a linear combination of the remaining vectors.

Definition 29.8. A set $\{v_1, v_2, \dots, v_n\}$ of vectors in a vector space V is called a **basis** for V if

- $\{v_1, v_2, \dots, v_n\}$ is linearly independent, and
- $\text{span}\{v_1, v_2, \dots, v_n\} = V$.

Theorem 29.9. Let $\{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V . Then every vector $v \in V$ can be written uniquely as a linear combination $v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$.

Theorem 29.10. Let V be a vector space that has a basis with n elements. Then any two bases of V have exactly n elements.

Definition 29.11. If $\{e_1, e_2, \dots, e_n\}$ is a basis for a vector space V , then we say the **dimension** of V is n and write $\dim V = n$. If V is the trivial vector space $\{\vec{0}\}$, we define $\dim V = 0$. A vector space is called **finite-dimensional** if it has a finite basis, and **infinite-dimensional** otherwise.

30 Lecture 29 – 4/21

30.1 21.1 Extension Fields

A field E is an **extension field** of a field F if F is a subfield of E . The field F is called the **base field**. We write $F \subset E$.

Theorem 30.1. Let F be a field and let $p(x)$ be a nonconstant polynomial in $F[x]$. Then there exists an extension field E of F and an element $\alpha \in E$ such that $p(\alpha) = 0$.

An element α in an extension field E over F is **algebraic** over F if $f(\alpha) = 0$ for some nonzero polynomial $f(x) \in F[x]$.

We say the extension field E is a finite extension of degree n over F if $E \cong F[x]/\langle p(x) \rangle$ for some irreducible polynomial $p(x)$ of degree n , denoted $[E : F] = n$.

An element in E that is not algebraic over F is **transcendental** over F .

30.2 21.2 Splitting Fields

Definition 30.2. Let F be a field and $p(x) = a_0 + a_1x + \cdots + a_nx^n$ be a nonconstant polynomial in $F[x]$. An extension field E of F is a **splitting field** of $p(x)$ if there exist elements $\alpha_1, \dots, \alpha_n$ in E such that $E = F(\alpha_1, \dots, \alpha_n)$ and $p(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n)$.

A polynomial $p(x) \in F[x]$ **splits** in E if it is the product of linear factors in $E[x]$.

Theorem 30.3. Let $p(x) \in F[x]$ be a nonconstant polynomial. Then there exists a splitting field E for $p(x)$.

Theorem 30.4. Let $p(x)$ be a polynomial in $F[x]$. Then there exists a splitting field K of $p(x)$ that is unique up to isomorphism.

31 Lecture 30 – 4/23

31.1 21.3 Geometric Constructions

A real number α is **constructible** if we can construct a line segment of length $|\alpha|$ in a finite number of steps from a segment of unit length by using a straightedge and compass.

Theorem 31.1. The set of all constructible real numbers forms a subfield F of the field of real numbers.

Lemma: If α is a constructible number, then $\sqrt{\alpha}$ is a constructible number.

Lemma: Let F be a field of constructible numbers. Then the points determined by the intersections of lines and circles in F lie in the field $F(\sqrt{\alpha})$ for some α in F .

Theorem 31.2. A real number α is a constructible number if and only if there exists a sequence of fields

$$\mathbb{Q} = F_0 \subset F_1 \subset F_2 \subset \cdots \subset F_k$$

such that $F_i = F_{i-1}(\sqrt{\alpha_i})$ with $\alpha_i \in F_{i-1}$ and $\alpha \in F_k$. In particular, there exists an integer $k > 0$ such that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^k$.

32 Lecture 31 – 4/28

32.1 22.1 Structure of Finite Fields

Definition 32.1. A field F has **characteristic** p if p is the smallest positive integer such that $p\alpha = 0$ for every nonzero element $\alpha \in F$. If no such integer exists, F has characteristic 0. It is known that p must be prime whenever the characteristic is finite.

Theorem 32.2. If F is a finite field, then the characteristic of F is p for some prime p .

Theorem 32.3. If F is a finite field of characteristic p , then $|F| = p^n$ for some $n \in \mathbb{N}$.

Definition 32.4. Let F be a field. A polynomial $f(x) \in F[x]$ of degree n is **separable** if it has n distinct roots in its splitting field, i.e. it factors into distinct linear factors over that splitting field. An extension E of F is a **separable extension** of F if every element of E is the root of some separable polynomial in $F[x]$.

Definition 32.5. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in F[x]$. The **derivative** of $f(x)$ is $f'(x) = a_1 + 2a_2x + \cdots + na_nx^{n-1}$. This is a formal derivative, and it is not defined in terms of limits as in calculus.

Theorem 32.6. Let F be a field and $f(x) \in F[x]$. Then $f(x)$ is separable if and only if $\gcd(f(x), f'(x)) = 1$.

Theorem 32.7. Let p be a prime and D an integral domain of characteristic p . Then for all $a, b \in D$ and all positive integers n , $(a + b)^{p^n} = a^{p^n} + b^{p^n}$.

Theorem 32.8. For every prime p and every positive integer n , there exists a finite field with p^n elements. Furthermore, any field of order p^n is isomorphic to the splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$.

Definition 32.9. The unique finite field with p^n elements (up to isomorphism) is called the **Galois field of order** p^n and is denoted $GF(p^n)$.

Theorem 32.10. Every subfield of $GF(p^n)$ has order p^m where $m \mid n$. Conversely, for each m with $m \mid n$ there exists a unique subfield of $GF(p^n)$ isomorphic to $GF(p^m)$.